Fund Risk Assessment Report

Input file: data/sample_nav_data.xlsx

Generated date: 2026-04-09

Metric	Value
Data Frequency	weekly
Annualization Factor	52
Rolling Windows	12, 26, 52
Inception Annualized Return	20.98%
Inception Return	84.65%
Annualized Volatility	31.11%
Maximum Drawdown	-41.73%
Maximum Drawdown Date	2024-10-18
Sharpe Ratio	0.6102
Sortino Ratio	1.0310
Calmar Ratio	0.5029

Data and Frequency Assessment

The report is based on 166 NAV observations from 2022-11-11 to 2026-01-30. The detected data frequency is weekly, so the analysis uses an annualization factor of 52.

Rolling metrics are calculated with frequency-aware windows: 12, 26, 52. This prevents daily assumptions from being applied to weekly or monthly data.

Return Performance Commentary

Since inception, the fund generated a total return of 84.65%, equivalent to an annualized return of 20.98% over the observed period.

The year-to-date return is -5.36%, while the trailing one-year return is 14.79%. The win rate is 50.30%, which measures the share of positive periodic return observations.

The return distribution skewness is 0.5154. Positive skewness suggests relatively more upside tail observations, while negative skewness would indicate more downside tail behavior.

Risk and Drawdown Commentary

Annualized volatility is 31.11%, which is classified as elevated based on the rule-of-thumb thresholds used in this report.

The maximum drawdown is -41.73%, occurring on 2024-10-18. This drawdown is classified as material and has not recovered within the available sample.

Drawdown analysis should be read together with the NAV chart and drawdown frequency chart, because maximum drawdown captures only the deepest loss while frequency captures how persistently the fund stayed below prior highs.

Tail Risk Commentary

The 95% 1-day equivalent VaR (scaled from weekly returns) is 2.65%, and the corresponding 10-day VaR is 8.37%. These figures estimate downside loss under historical simulation assumptions.

The 99% 1-day equivalent VaR (scaled from weekly returns) is 2.65%, and the corresponding 10-day VaR is 8.37%. The 99% confidence level focuses on more severe tail observations.

The 95% 1-day equivalent CVaR/ES (scaled from weekly returns) is 2.65%, while the 99% 1-day equivalent CVaR/ES (scaled from weekly returns) is 2.65%. CVaR/ES estimates the average loss within the tail, so it is useful for understanding loss severity beyond the VaR threshold.

Note: For non-daily data, short-horizon VaR and CVaR are scaled estimates based on the underlying return frequency.

Risk-Adjusted Return Commentary

The Sharpe ratio is 0.6102, which is classified as acceptable. The Sortino ratio is 1.0310, which focuses on downside volatility rather than total volatility.

The Calmar ratio is 0.5029, linking annualized return to maximum drawdown. A lower Calmar ratio suggests that return generation has come with a larger path-dependent downside burden.

The Treynor ratio is NaN and the Treynor-Black ratio is NaN. These values are NaN when the input data does not include a benchmark_return column.

Metrics Summary

Category	Metric	Value
Data Profile	input_file	data/sample_nav_data.xlsx
Data Profile	start_date	2022-11-11
Data Profile	end_date	2026-01-30
Data Profile	observations	166.0000
Data Profile	data_frequency	weekly
Data Profile	annualization_factor	52
Data Profile	rolling_windows	12, 26, 52
Return Metrics	inception_annualized_return	20.98%
Return Metrics	inception_return	84.65%
Return Metrics	year_to_date_return	-5.36%
Return Metrics	one_year_return	14.79%
Return Metrics	win_rate	50.30%
Return Metrics	return_skewness	0.5154
Return Metrics	current_consecutive_up_months	0
Return Metrics	current_consecutive_down_months	1
Return Metrics	max_consecutive_up_months	6
Return Metrics	max_consecutive_down_months	3
Risk Metrics	annualized_return	20.98%
Risk Metrics	annualized_volatility	31.11%
Risk Metrics	max_drawdown	-41.73%
Risk Metrics	max_drawdown_date	2024-10-18
Risk Metrics	max_drawdown_recovery_days	NaN
Tail Risk Metrics	var_95_1d_scaled	2.65%
Tail Risk Metrics	var_95_10d	8.37%

Metrics Summary

Category	Metric	Value
Tail Risk Metrics	var_99_1d_scaled	2.65%
Tail Risk Metrics	var_99_10d	8.37%
Tail Risk Metrics	cvar_es_95_1d_scaled	2.65%
Tail Risk Metrics	cvar_es_95_10d	8.37%
Tail Risk Metrics	cvar_es_99_1d_scaled	2.65%
Tail Risk Metrics	cvar_es_99_10d	8.37%
Risk-Adjusted Return Metrics	sharpe_ratio	0.6102
Risk-Adjusted Return Metrics	sortino_ratio	1.0310
Risk-Adjusted Return Metrics	calmar_ratio	0.5029
Risk-Adjusted Return Metrics	treynor_ratio	NaN
Risk-Adjusted Return Metrics	treynor_black_ratio	NaN
Risk Signals	vol_signal	high
Risk Signals	drawdown_signal	deep
Risk Signals	sharpe_signal	average
Risk Signals	tail_signal	low
Risk Signals	trend_signal	stable
Risk Signals	overall_risk	High
Risk Signals	key_risk_driver	Drawdown
Risk Signals	monitoring_flag	Escalate
Risk Narrative	risk_narrative	The fund exhibits elevated volatility, with sig

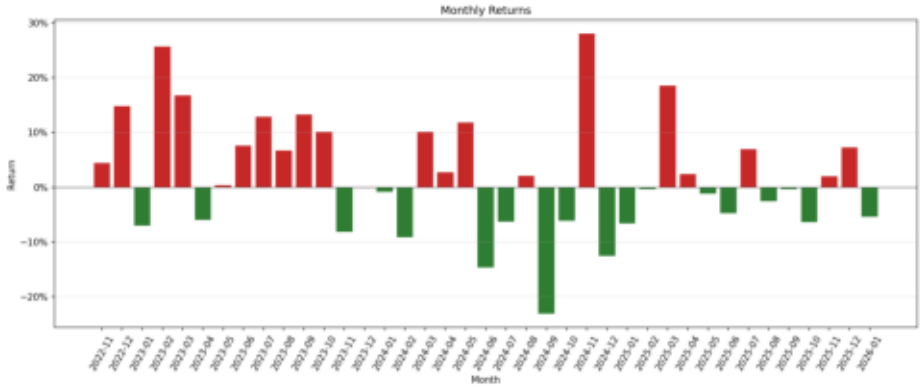
Risk Commentary

The fund exhibits elevated volatility, with significant drawdown pressure. The fund shows moderate risk-adjusted performance, while risk conditions remain stable. Overall, the portfolio is assessed as High risk, primarily driven by drawdown pressure. Monitoring status: Escalate.

Return Analysis

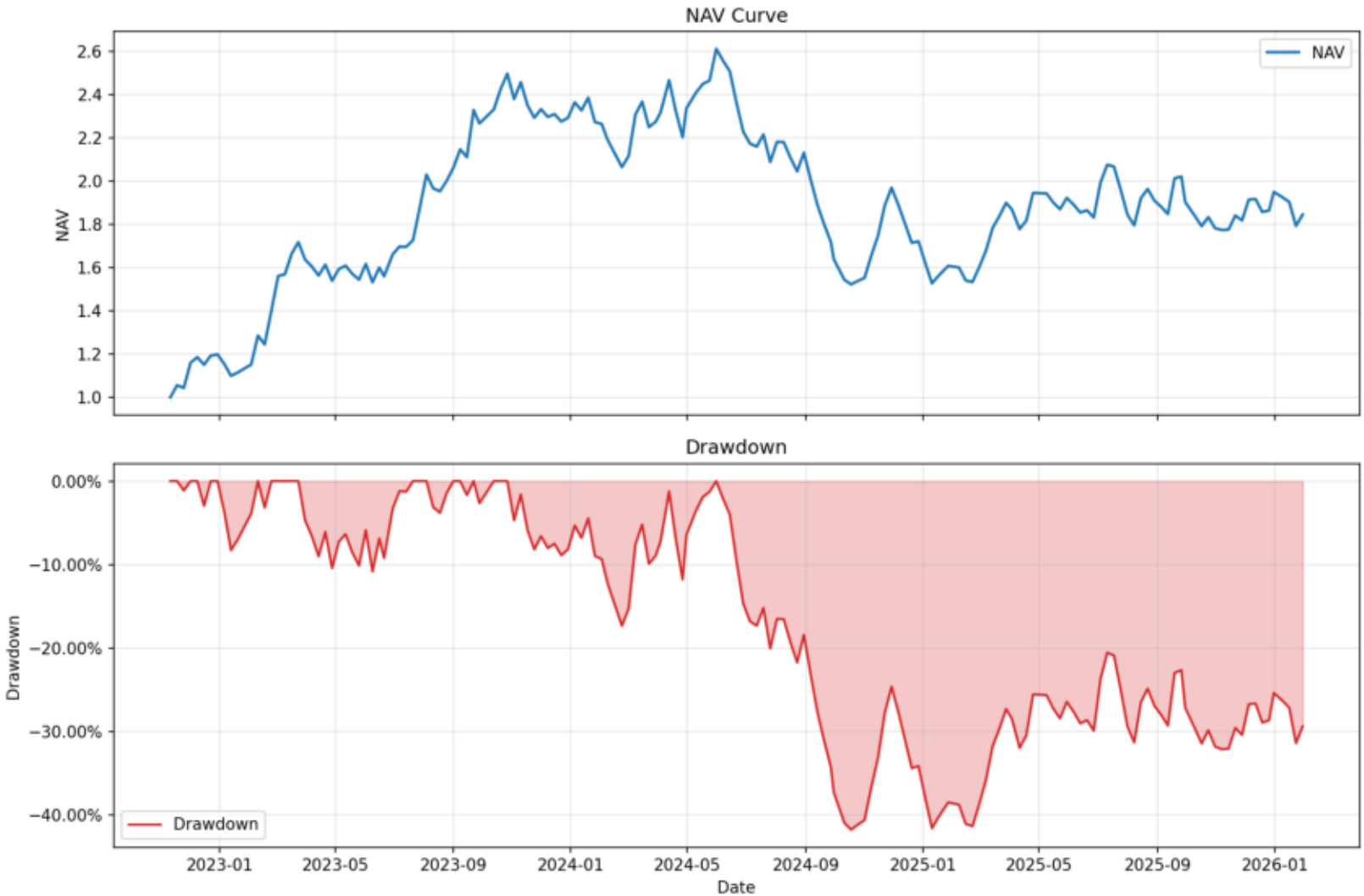
Annual Return

Monthly Return

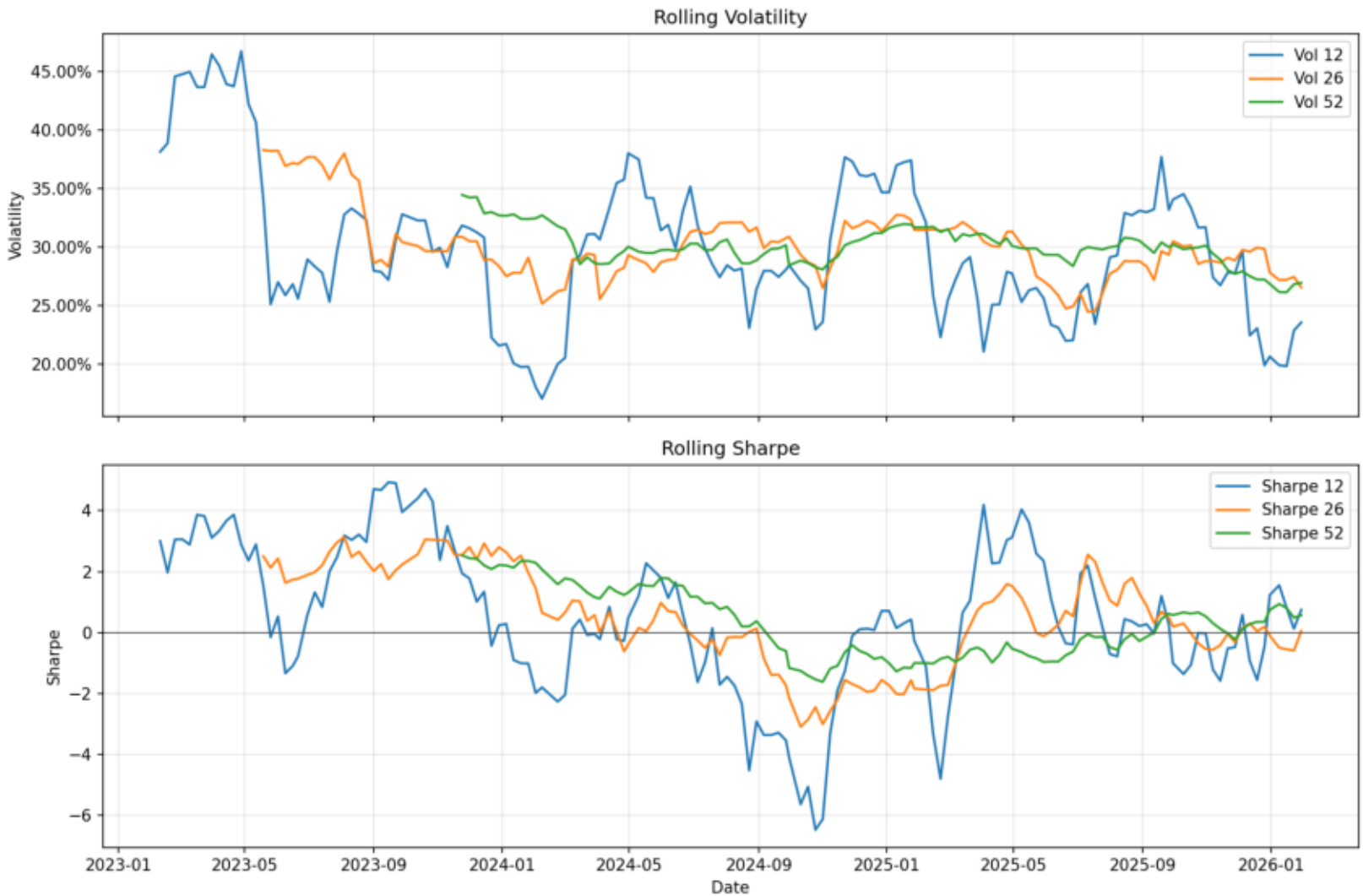


Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual Retur	Win Rate
2022											4.40%	14.84%	**19.89%**	**100.00%**
2023	-6.99%	25.76%	16.79%	-5.99%	0.33%	7.62%	12.88%	6.69%	13.29%	10.15%	-8.17%	0.01%	**91.40%**	**75.00%**
2024	-0.88%	-9.17%	10.13%	2.76%	11.79%	-14.68%	-6.29%	2.05%	-23.10%	-6.16%	28.05%	-12.59%	**-24.94%**	**41.67%**
2025	-6.61%	-0.38%	18.61%	2.37%	-1.14%	-4.72%	6.96%	-2.53%	-0.33%	-6.36%	2.02%	7.24%	**13.28%**	**41.67%**
2026	-5.36%												**-5.36%**	**0.00%**

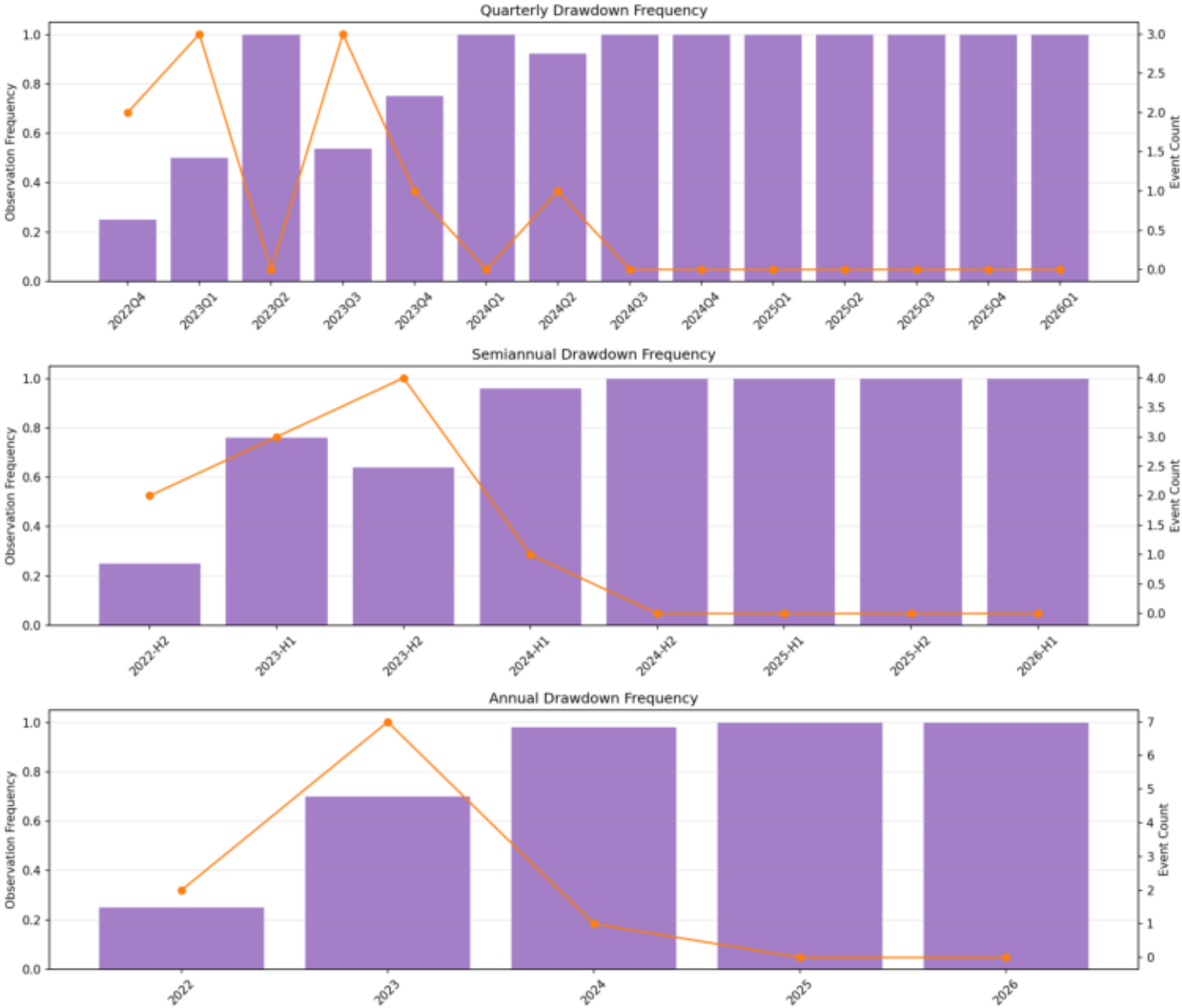
NAV and Drawdown



Rolling Risk Metrics



Drawdown Frequency



Conclusion and Risk Notes

Overall, this fund demonstrates a historical risk-return profile that should be interpreted through both absolute performance and path-dependent downside behavior.

From a historical perspective, the fund's risk-adjusted profile appears less robust, and further investigation into downside drivers, market sensitivity, and recovery dynamics would be advisable.

Annualized volatility is 31.11%, so ongoing monitoring should focus on whether future realized volatility remains consistent with the historical pattern.

This report is based solely on historical NAV behavior and should be supplemented with strategy-level due diligence, exposure decomposition, and qualitative review before forming any investment decision.